Online Appendix
"Heterogeneous Spillovers of Housing Credit Policy" *

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*The views expressed in this paper are those of the author and do not necessarily represent the views of the Bank of Spain and the Eurosystem.Remaining errors are the authors'.

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A Agency and Market Data

Below we describe the macroeconomic data used in the empirical section of the paper. We follow Fieldhouse, Mertens and Ravn (2018) closely in constructing the variables that we use.[1] *Residential mortgage debt* is the sum of home mortgages and multifamily residential mortgages from the Federal Reserve's Financial Accounts of the United States. *Nominal GDP* is from the National Income and Product Accounts. *Agency mortgage holdings* is the sum of the retained mortgage portfolios of Fannie Mae and Freddie Mac. Between 1980 and 2003, the data on retained mortgage portfolio is available from various issues of Federal Reserve Bulletin. After 2003 the data is from Fannie Mae's and Freddie Mac's monthly volume summaries combined with annual OFHEO/FHFA reports.[2] *Residential mortgage originations* before 1997 is from monthly releases of the Survey of Mortgage Lending Activity from the HUD. After 1997 the data on originations is available from Datastream (series USMORTORA) via The University of Manchester . *Net portfolio purchases* is the sum of corresponding series for Fannie Mae and Freddie Mac. Individual series before 2003 are available from various issues of Federal Reserve Bulletin. After 2003 the data is from Fannie Mae's and Freddie Mac's monthly volume summaries. *Conventional mortgage rate* is the 30-year fixed-rate conventional conforming mortgage rate, available at Freddie Mac mortgage market survey. *Housing starts* is obtained from FRED database at the Federal Reserve Bank of St. Louis (series HOUST). *House prices* is measured by the Freddie Mac house price index (FMHPI) available on Freddie Mac's website. *Nominal price level* is obtained from FRED database at the Federal Reserve Bank of St. Louis (series PCEPILFE). *Personal income* is obtained from FRED database at the Federal Reserve Bank of St. Louis (series PI). *Unemployment rate* is obtained from FRED database at the Federal Reserve Bank of St. Louis (series UNR). *Short- and long-term interest rates* are 3-month and 10-year Treasury rates, obtained from FRED database at the Federal Reserve Bank of St. Louis (series TB3MS and GS10). *BAA and AAA corporate bond rates* are the Moody's seasoned BAA and

[1] Please note that during the writing of this paper, the data and replication materials from Fieldhouse, Mertens and Ravn (2018) were not yet publicly available and all the variables described in the appendix are own calculations. As the data for Ginnie May was not available, it is excluded from the definitions of agency mortgage holdings and agency purchases. Similarly, net commitments were not publicly available for all years and were thus excluded. Other variables follow closely those of Fieldhouse, Mertens and Ravn (2018).

[2] Freddie Mac's monthly volume summaries that were not available online or via U.S. Securities and Exchange Commission for the years 2003-2008 were kindly provided by Freddie Mac's Equity Relations Department.

AAA yields, obtained from FRED database at the Federal Reserve Bank of St. Louis (series BAA and AAA).

B Testing for Exclusion Restrictions

Below we decribe how we choose the horizon of cumulative agency net purchases. We start by running regression

$$\frac{\sum_{j=0}^{h} p_{t+j}}{X_t} = \tilde{a}_h + \tilde{c}_h \frac{\tilde{m}_t}{X_t} + \tilde{d}_h(L) Z_{t-1} + \tilde{u}_{t+h}, \tag{1}$$

for horizons $h = 1$ (one horizon) to $h = 20$ (five years) and calculate the robust F-statistics of exluding $\tilde{m}_t$. Figure B.1 plots the value of this statistics for all the horizons. As Figure B.1 displays, the F-statistics exceeds value 10 for horizons between 1 and 3 quarters (indicating that narrative measure $\tilde{m}_t$ is a strong predictor for agency purchases for this horizon), and reaches the highest value of around 15 when $h = 2$. The the robust F-statistics declines for longer horizons. [3]

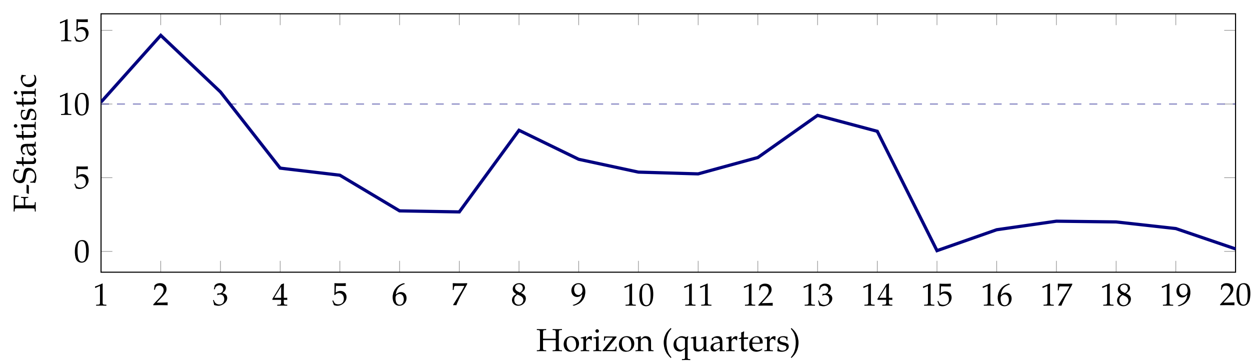

Figure B.1: **First Stage Robust F-statistic.** Figure displays robust F-statistics on the excluded instrument of the first-stage regressions of cumulative agency net purchases. Horizontal dashed line represents the threshold level of 10.

As such, for our first-stage estimation we pick $h = 2$, and use the fitted values of $\frac{\sum_{j=0}^{2} p_{t+j}}{X_t}$ in our second-stage regression (**??**). The interpretation of our second-state estimates in equation (**??**) is then the following: we will analyze the response of the variable of interest y_t to and increase in agency purchases that is anticipated 2 quarter ahead.

[3] Using monthly data, Fieldhouse, Mertens and Ravn (2018) also find that narrative measure is a strong predictor for the horizons between 4 to 48 months, reaching the highest value at 8 months. While our estimates of F-statistics decline faster for longer horizons, the results are in line with Fieldhouse, Mertens and Ravn (2018) when measuring the peak of predictability at around quarter 2.

C Extra Empirical Results

C.1 Tenure Housing Status Switchers

Our empirical analysis in section ??, that relies on employing a grouping estimator in the spirit of Browning, Deaton and Irish (1985), rests on the assumption that households do not endogenously switch between different categories because of the changes in agency purchases. To check if this is indeed the case, we first check how many households in the CEX change their housing tenure status (to asses the importance of those switchers). We then check if our agency purchase shocks lead to endogenous switching between the housing tenure status.

Switchers in the CEX Being a rotating panel, CEX follows households up to 4 consecutive quarters, and collects the information on housing tenure status each quarter. To check how many households change the housing tenure status between interviews, we calculate and plot (see Figure C.1) the number of households that change the status between the current and previous quarter (blue line) as well as between the current quarter and two quarters before (red line). As the Figure reports, the number of households that change housing tenure status varies between 4 and 150 for our sample period.

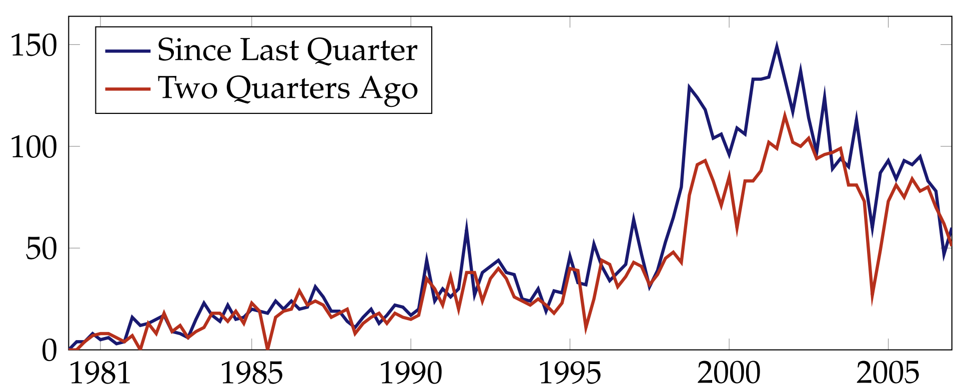

Figure C.1: Number of households that change the status between the current and previous quarter.

To identify what share of households this number of switchers represent, Figure C.2 below plots the share of households that switch status between consecutive quarters (blue line) and between two quarters (red line) as share of total households in the given quarter in the CEX.

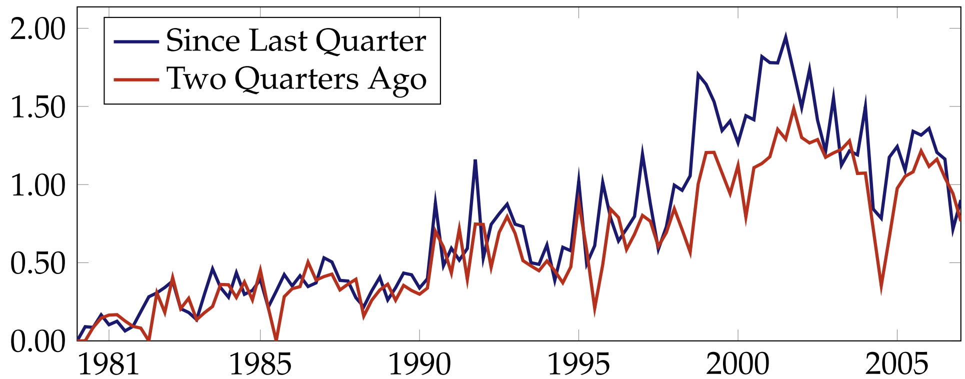

Figure C.2: Share of households that switch status between consecutive quarters.

As Figure C.2 reports, the share of households that switch housing tenure status in the data represents, on average, 0.7% of the sample.

Testing for Endogenous Switching Above we reported that switchers represent only 07% of the overall sample of households in the CEX. Yet, we need to assure that there is no endogenous switching between three different groups of households that we identify. As such, we construct several indicator variables that identify several possibilities of switching. Those are renters becoming mortgagors, renters becoming outright homeowners, mortgagors becoming homeowners, and outright owners becoming mortgagors.

We then run regression (**??**) with the dependent variable being number of switchers in the current quarter. Impulse response functions are displayed in Figure C.3. As the Figure reports, the change in switchers due to the increase in agency purchases is tiny and insignificant, implying that our grouping estimator is valid in this context.

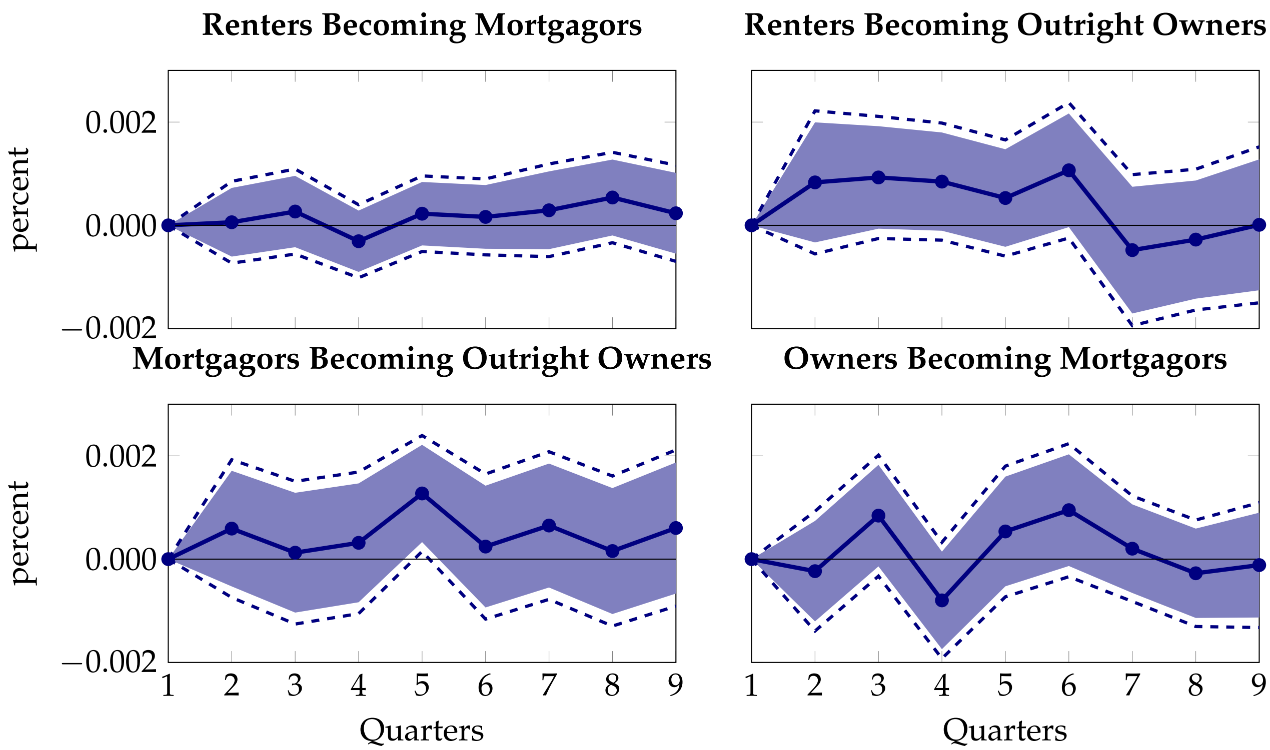

Figure C.3: Impulse response of number of switchers to a 1% increase in net purchase by FNMA & FHLMC, anticipated 2 quarters before. Blue areas and broken lines represent 90% and 95% confidence intervals, respectively.

C.2 Other Consumption Responses: Age and Housing Tenure Status

In this section we extend the analysis of Section 2 to also analyze the impulse of response of non-durable expenditure when we interact both the age and the housing tenure status. The results are in line with those reported in Section 2: among mortgagors, the response is large for all age groups, but is significant for younger households. Among renters, the largest response in expenditure is for younger renters. It is worth noting, that for older mortgagors and renters, as well as for young owners, the results are noisy and highly insignificant, given that those groups represent small part of the overall sample.

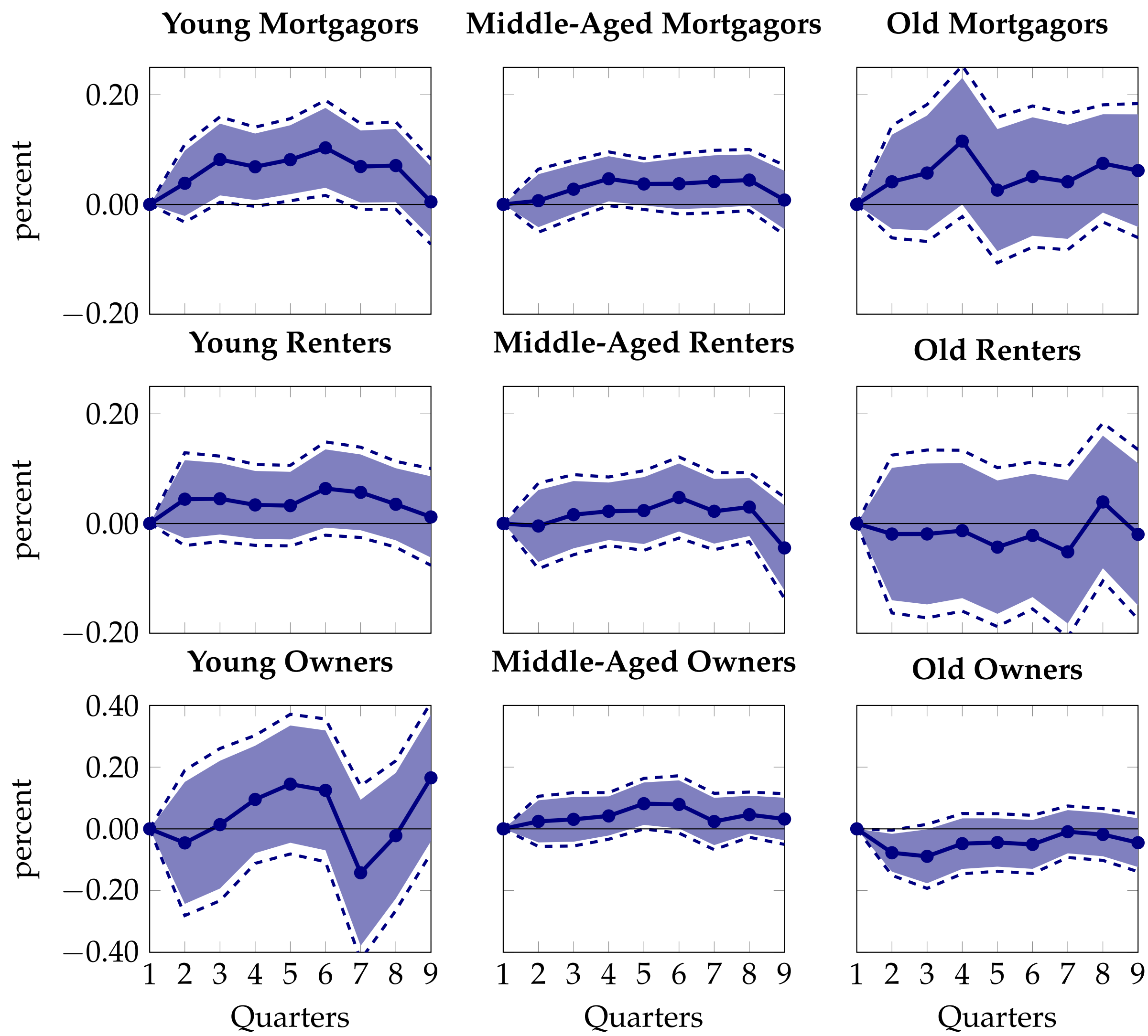

Figure C.4: Impulse response of non-durable expenditure a 1% increase in net purchase by FNMA & FHLMC, anticipated 2 quarters before. Blue areas and broken lines represent 90% and 95% confidence intervals, respectively.

C.3 Other Consumption Responses: Durable Expenditure

In this section we extend the analysis of Section 2 to also look at the response of durable expenditure following an increase in agency purchases. There are several issues worth noting before we present the results. Firstly, in terms of measurements, mapping durable expenditure to durable consumption is not straightforward, as durable goods provide services for several periods (unlike non-durable consumption, that maps almost directly to expenditure). Moreover, to reconstruct the durable consumption of the household one would need to know the value of the current stock of durables, the information that is missing from the CEX. Of course, expenditures of durables could still be an important aspect in the context of the housing tenure dimension. For example, Benmelech, Guren and Melzer (2022) report that new homeowners increase their expenditure in durables in the two years following a purchase of a home. Unfortunately, given that the CEX follows households only for 4 quarters, this channel is not something that would be seen in the empirical section. Finally, in terms of how the empirical results map into the structural model, introducing durable consumption into the model would involve expanding both the choice set (non-durable vs durable expenditure) as well as the state space (stock of durables) into an already complicated model. As such, the impulse responses below play more of an indicative role.

We start with looking at the response of durable expenditures for all households following an increase in agency purchases. As Figure C.5 indicates, durable expenditure does not seem to change following the shock.

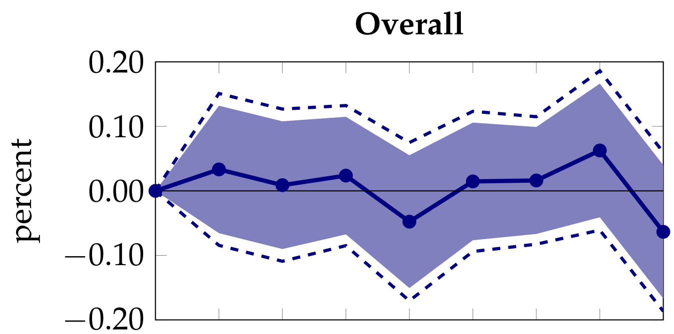

Figure C.5: Impulse response of total durable expenditure a 1% increase in net purchase by FNMA & FHLMC, anticipated 2 quarters before. Blue areas and broken lines represent 90% and 95% confidence intervals, respectively.

Once we divide households based on their housing tenure status, as Figure C.6 indicates, the flat response in Figure C.5 is a combination of a decrease (even though insignificant) of renters and an increase in durable expenditures for outright homeowners. Unfortunately, given the data available we cannot directly link the increase in durable expenditure we find to Benmelech, Guren and Melzer (2022), as we do not directly observe the adjustment of expenditures, savings and debt for this group.

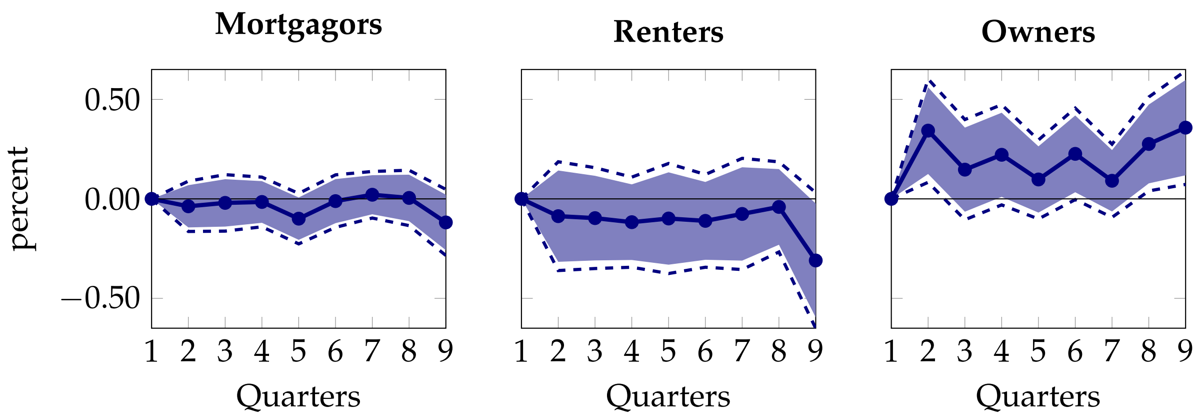

Figure C.6: Impulse response of durable expenditure by housing tenure status a 1% increase in net purchase by FNMA & FHLMC, anticipated 2 quarters before. Blue areas and broken lines represent 90% and 95% confidence intervals, respectively.

Finally, Figure C.7 repeats the analysis for groups of households of different ages. Given the large standard errors of the estimated impulse responses it is hard to reach a concrete conclusion, but the results seem consistent that older households (who are more likely to be homeowners) increase expenditure on durable goods the most.

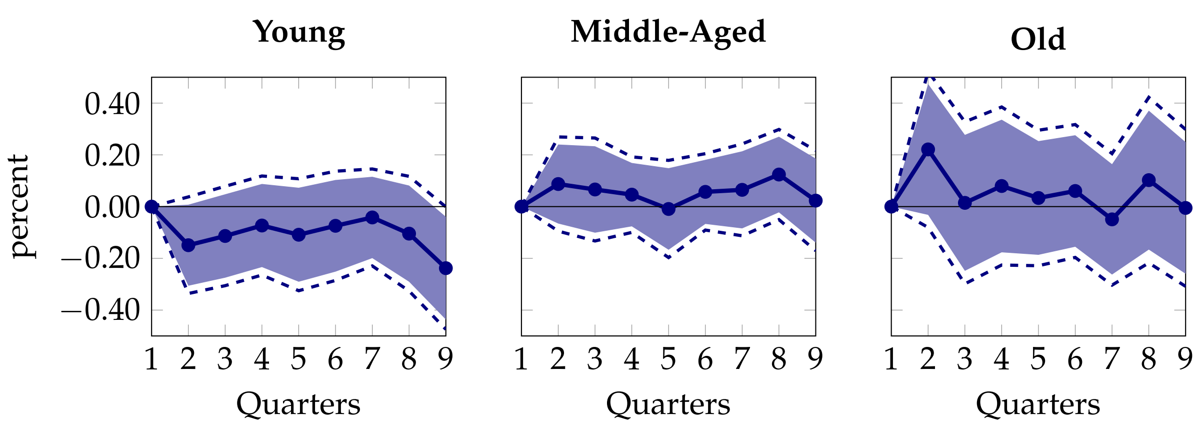

Figure C.7: Impulse response of durable expenditure by age a 1% increase in net purchase by FNMA & FHLMC, anticipated 2 quarters before. Blue areas and broken lines represent 90% and 95% confidence intervals, respectively.

D Definition of Equilibrium and Computational Appendix

In this section we describe the definition of equilibrium in the structural model, as well as provide some details on the numerical solution and the simulation of the model.

D.1 Definition of Equilibrium

Our definition of equilibrium consists of households' consumption decision rules $\{c^r(\mathbf{x}^n), c^o(\mathbf{x}^n), c^p(\mathbf{x}^h), c^f(\mathbf{x}^h), c^s(\mathbf{x}^h)\}$, savings decision rules $\{a^r(\mathbf{x}^n), a^o(\mathbf{x}^n), a^p(\mathbf{x}^h), a^f(\mathbf{x}^h), a^s(\mathbf{x}^h)\}$, mortgage decision rules $\{m^o(\mathbf{x}^n), m^f(\mathbf{x}^h), \pi(\mathbf{x}^h)\}$, and housing choice rules $\{\tilde{h}^r(\mathbf{x}^n), h^o(\mathbf{x}^n), h^p(\mathbf{x}^h), h^f(\mathbf{x}^h), \tilde{h}^s(\mathbf{x}^h)\}$, and government expenditure G, such that

1. Households' policy function solve problems (**??**), (**??**), (**??**), (**??**) and (**??**) given prices p_h and ρ_h
2. Government expenditure G clears governmental budget constraint
3. Supply of owner-occupied housing is fixed at some level $\widehat{H}$. The price of housing p_h is such that in equilibrium, the demand for owner-occupied housing (which consists of new houses bought, existing housing of mortgage payer minus housing of sellers) is equal to the supply. Similarly, ρ_h is determined by the equilibrium conditions in the rental market.

D.2 Numerical Solution Details

We solve the model using Value Function Iteration via backward induction starting with the final period of life. We discretize the state space by fixing grids for income, owner house size, mortgage and assets. To facilitate the solution algorithm, both asset and mortgage choices, as well as housing choices for both renters and homeowners, are restricted to be on grid. In our solution algorithm, we fix the size of income grid to have 7 points, housing to 4 points, mortgages to 35 points and assets to 30 points. In total, we solve 5 different value function maximization problems simultaneously: those of renters that remain renters, renters who decide to become homeowners, homeowners who choose the mortgage payments, homeowners that sell the house, as well as homeowners who decided to refinance.

D.3 Model Simulation Details

With optimal policy functions available, we simulate the deterministic steady-state of the model iterating forward on the policy functions. We start with initial 50000 households in age 21 (that corresponds to age 1 in the model), starting with the initial distribution of income, assets, housing and mortgages, and iterate forward on those households. While the model allows for exogenous probability of early death, in practice we keep the size of each cohort/age group fixed throughout, and instead adjust the cohort weight to reflect the probability of survival when calculating the moments. To initiate the initial joint distribution over income, assets, mortgages and housing, we assume that all households start with zero housing, assets or debt, and we sample initial income state based on the invariant distribution. When simulating transitional dynamics, due to memory constraints we decrease the number of simulated households to 6000 households per age and period.

E Extra Details of Parametrization

E.1 Life-Cycle Profile Estimation

We use the 2001 wave of the Survey of Consumer Finance (SCF) to extract the life-cycle component of earnings (that corresponds to χ_j in the model). For that, we use the total labor earnings of the households in the SCF (variable *X5702*). We regress the log of total labor earnings (normalized by $50,000) on 4th order polynomial of age, for households whose age of head is between 21 and 65 years old, that is we estimate the coefficients of

$$\chi_j = a_0 + a_1 j + a_2 j^2 + a_3 j^3 + a_4 j^4, \quad \text{where} \quad j \in \{1, \cdots, J_r\}$$

The estimates are reported in the Table E.1 below.

Estimates of Life-Cycle Component of Income		
Coefficient	Value	St. Error
a_0	-1.4211	(0.0678)
a_1	0.2091	(0.0189)
a_2	-0.0132	(0.0016)
a_3	0.0004	(5.54E-05)
a_4	-4.38E-06	(6.24E-07)

Table E.1: Estimates of life-cycle component of labor income from the 2001 wave of the SCF. Standard errors are bootstrap standard error based on 100 replications.

E.2 Estimating Equivalence Scale

To construct the exogenous equivalence scale e_j we use the pooled waves of the CEX between 1980 and 2007 to calculate the OECD equivalence scale. The scale assigns value of 1 to the head of household, value of 0.7 to each additional adult and value of 0.5 for each child. We then regress this scale on 4th order polynomial of age, for households whose age of head is between 21 and 80 years old. That is, we estimate

$$e_j = a_0^e + a_1^e j + a_2^e j^2 + a_3^e j^3 + a_4^e j^4, \quad \text{where} \quad j \in \{1, \cdots, J\}$$

The estimates are reported in the Table E.2 below.

Estimates of Equivalence Scale	
Coefficient	Value
a_0^e	1.4722
a_1^e	0.1304
a_2^e	-0.0048
a_3^e	5.61E-05
a_4^e	-1.67E-07

Table E.2: Estimates of equivalence scale from the 1980-2007 waves of the CEX.

E.3 Survival Probabilities

We use the 2018 Period Life Tables from the United States Mortality Database to construct survival probabilities s_j's. Those are reported in Table E.3. We utilized survival probabilities for both genders across the entire United States. Since the maximum age is $J = 60$ in the model (which corresponds to age 80 in real life), we impose that $s_{J+1} = 0$.

E.4 Some definitions from the SCF

We use 2001 wave of the Survey of Consumer Finances (SCF) to construct the moments used in the paper (both targeted and non-targeted) as well as the life-cycle profiles. The total housing wealth is constructed as the total sum of all residential real estate owned by households in the SCF (variables HOUSES and ORESRE). Mortgage debt is measured as debt secured by primary or other residential properties (sum of variables MRTHEL + HELOC + OTHLOC + RESDBT). Liquid wealth is defined as the sum of deposits in the transaction accounts (LIQ).[4]. The total net worth is defined as sum of housing and liquid wealth net of total debt. Total household labor earnings (variable X5702) is taken to represent household income in the model.

[4]We refer the reader to Federal Reserve's Survey of Consumer Finances website `https://www.federalreserve.gov/econres/scfindex.htm` for the exact definitions of those variables

Age (j)	s_{j+1}	**Age (j)**	s_{j+1}
1	0.9990	31	0.9956
2	0.9990	32	0.9951
3	0.9990	33	0.9946
4	0.9989	34	0.9942
5	0.9989	35	0.9937
6	0.9989	36	0.9933
7	0.9988	37	0.9926
8	0.9987	38	0.9921
9	0.9988	39	0.9914
10	0.9987	40	0.9910
11	0.9986	41	0.9903
12	0.9986	42	0.9895
13	0.9986	43	0.9889
14	0.9985	44	0.9880
15	0.9984	45	0.9873
16	0.9984	46	0.9867
17	0.9983	47	0.9855
18	0.9982	48	0.9845
19	0.9981	49	0.9821
20	0.9981	50	0.9826
21	0.9980	51	0.9799
22	0.9979	52	0.9780
23	0.9977	53	0.9751
24	0.9975	54	0.9738
25	0.9973	55	0.9710
26	0.9971	56	0.9688
27	0.9970	57	0.9654
28	0.9966	58	0.9620
29	0.9963	59	0.9582
30	0.9959	60	0.9534

Table E.3: Survival Probabilities from the 2018 USMD

F Allowing for House Price Changes

Fieldhouse, Mertens and Ravn (2018) find that following an increase in agency mortgage purchases, house prices rise gradually but very persistently over time, but this increase in house prices is relatively small and imprecisely estimated (see Figure F.1 that reproduces Figure X in Fieldhouse, Mertens and Ravn (2018)). Below we repeat the main analysis where we also exogenously change house prices (together with interest rates) as a result of an increase in agency purchases.

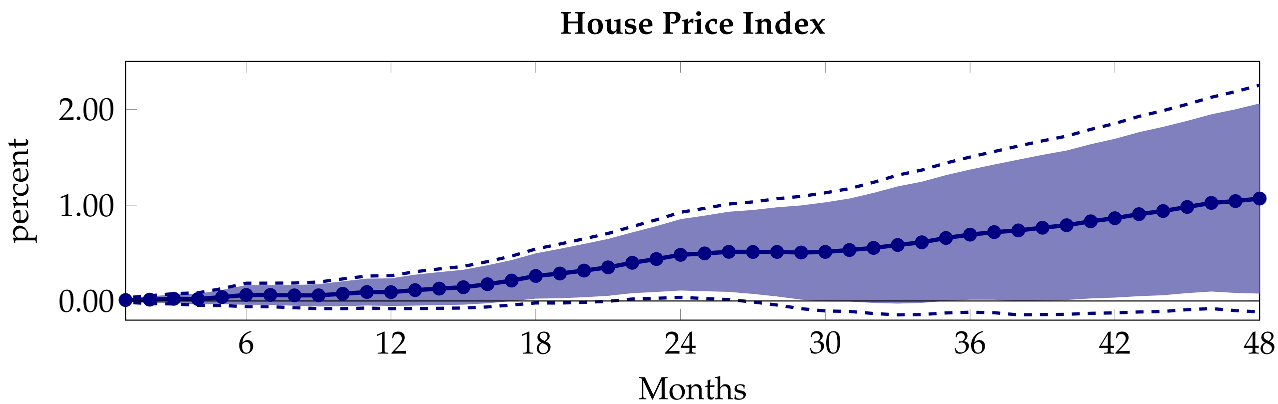

Figure F.1: This figure reproduces Figure X in Fieldhouse, Mertens and Ravn (2018). Blue areas and broken lines represent 90% and 95% confidence intervals, respectively.

To map the increase in house prices documented in Figure F.1 into the model. we treat the increase in house prices as permanent shock, that gradually increases to a value that is 1% higher than the initial steady state. Figure F.2 plots the path of house prices that we feed into the model. As in the steady state, the rental rate is assumed to be a constant fraction of house price, we assume that rental rate responds in a similar magnitude.

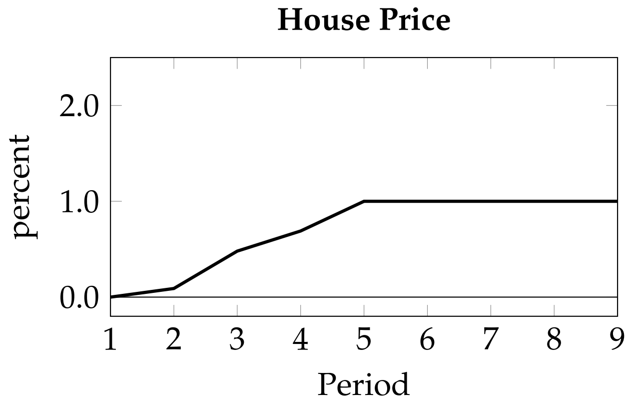

Figure F.2: The figure plots the path of house price that we feed into the structural model.

With now three shocks in hand (a temporary increase mortgage rate and interest rate, as well as permanent increase in house prices), we simulate the model forward and compare the implied path of consumption in this alternative economy with the one from Section **??**. A note of caution on the interpretation of the results: in the simulation exercise we assume that all shocks are unexpected, but once they are realized - households have a perfect foresight of their path. In the context of this experiment it implies that households know that the effect on interest and mortgage rate is short-lived, while the increase in house prices will remain forever.

As before, we start at looking at the response in expenditure for all households. Figure F.3 plots the simulated path of expenditure in the original experiment (black line) to that of alternative one (red line). As Figure illustrates, households still increase their expenditure following a decrease in the mortgage and interest rates, however as the price increases the expenditure converges to a new steady-state level that is in fact lower than the initial one.

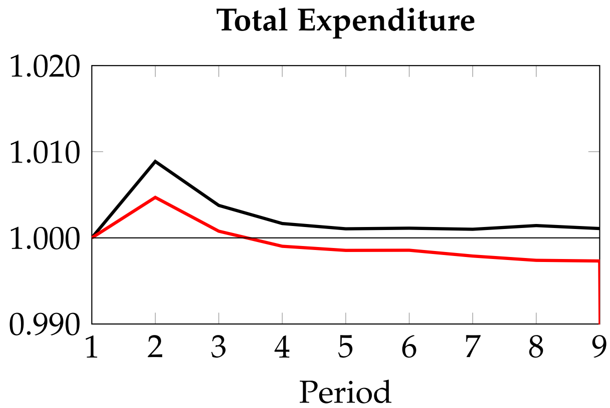

Figure F.3: The figure plots the simulated path of average expenditure for all households. Initial period is normalized to 1.

Again, to understand what drives the results presented above, we start at looking at different groups of households. First, we look at the path of simulated expenditure for households based on their housing tenure status. Top panel in Figure F.4 plots the comparison for mortgagors, middle panel for renters while the bottom panel to that of outright homeowners. Black line corresponds to the simulated path from the original experiment, while red line corresponds to alternative one. The simulated path of expenditure for mortgagors now has two different peaks - the initial one, similar to the original experiment when the mortgage rates are the lowest, as well as the second one that corresponds to the new peak of house prices. Indeed, as top panel of Figure F.4 also illustrates, households who refinance in the period when house prices reach the peak increase their consumption more than the households who do not refinance in that period. As we discussed in Section **??**, when refinancing households have two possible sources of extra income - one coming from the lower payments and one from relaxing the payment to income constraint. In fact, when the house prices are changing, the third source of possible income is the relaxation of the LTV constraint. When the value of the house is higher, households have now more access to credit then in the fixed price scenario.

Interestingly, when the shocks to interest rates and mortgage rate is accompanied by an increase in house prices, renters, compared to the baseline scenario, now increase their consumption for several periods. Simulations indicate then when faced with high future price of housing, the lifetime benefit of owning the house now decreases for the current renters, and lower interest rates result in lower savings and higher consumption. Again, this points to an important behavioral dynamic problem that households face. Moreover, portion of renters includes households who decide to sell the house. While facing the higher house prices and lower interest rates, those households have an extra cash flow that they use towards consumption.

Finally, in the model, for the outright homeowners, higher house prices implies higher housing maintenance expenditure (the cost is proportional to the value of the house). As such, over time they decrease their consumption towards a lower steady-state value.

Figure F.4: The figure plots the simulated path of average expenditure for households based on the housing tenure status. Top panel is for mortgagors, middle panel is for renters, while bottom panel is for outright owners. Black line corresponds to the original experiment in Section 5, red line corresponds to the alternative experiment in the current section. Initial period is normalized to 1.

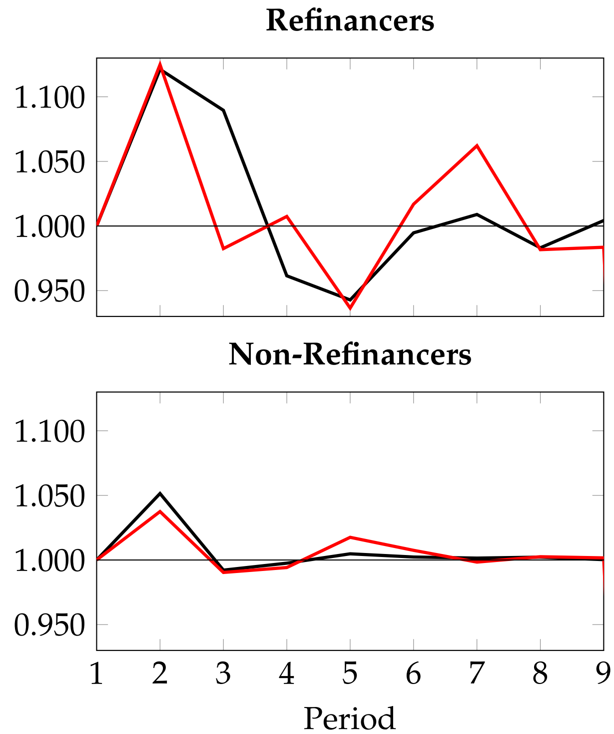

Figure F.5: The figure plots the simulated path of average expenditure for households based on the refinancing decision. Top panel is for households that refinance, while bottom panel is for households that do not. Black line corresponds to the original experiment in Section 5, red line corresponds to the alternative experiment in the current section. Initial period is normalized to 1.

Finally, we look at the path of simulated expenditure for households based on their age. Top panel in Figure F.4 plots the comparison for young households, middle panel for middle-aged, while the bottom panel to that of older households. As before, black line corresponds to the simulated path from the original experiment, while red line corresponds to alternative one. As the Figure indicates, the initial increase in expenditure due to the decrease in mortgage and interest rates is now muted, followed by an overall decrease in expenditure at the peak of house price increase. This patters is mostly pronounced for young and middle-aged households, with older households now only slightly increasing the expenditure.

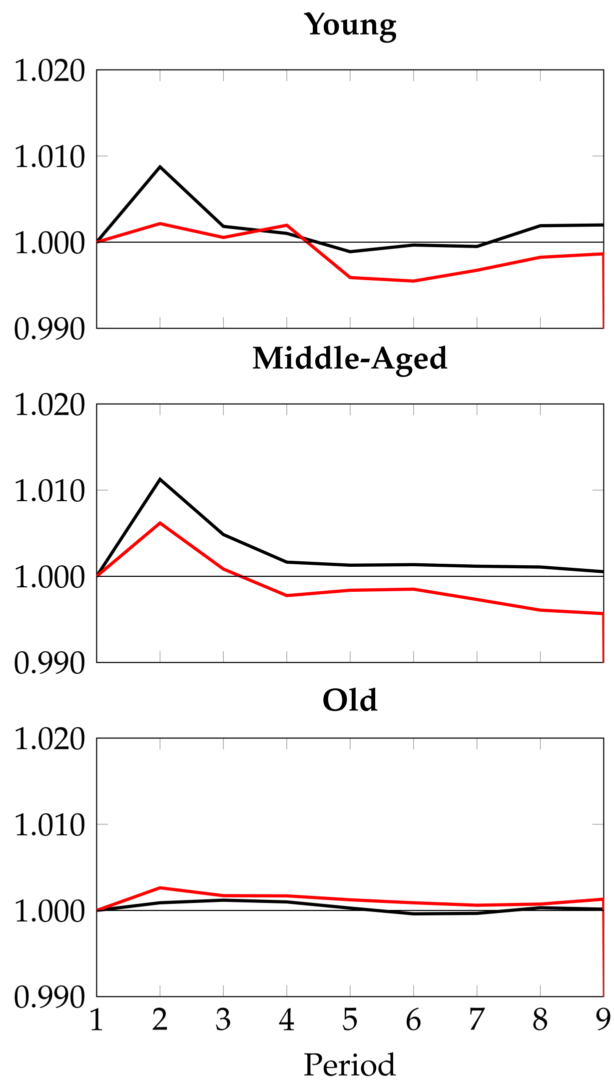

Figure F.6: The figure plots the simulated path of average expenditure for households based on the households age. Top panel is for young households, middle panel is for middle-aged, while bottom panel is for older. Black line corresponds to the original experiment in Section 5, red line corresponds to the alternative experiment in the current section. Initial period is normalized to 1.

G Non-Accommodative Monetary Policy

As mentioned in Section 5, empirically, expansionary credit policy results in both a drop in interest and mortgage rates (what Fieldhouse, Mertens and Ravn (2018) call an evidence of the "accommodative monetary policy" that follows credit policy changes). In this Appendix we analyze how important is the role of accomodative monetary policy for propagation of expansionary credit policy changes. In particular, we analyze the response of consumption to credit policy changes while keeping short-term interest rate fixed and only allowing the spread between the rates to move. That is, we allow the mortgage rate to change according to right panel of Figure 7. The implied change in the mortgage rates is more drastic (even thought it is smaller) then in the original experiment, as the change in spread becomes close to zero already in the second period.[5] Given that the model is solved and simulated on a discrete grid, for the simulations, this implies that this smaller and short-lived shock will generate more noise relative to a larger, more gradual change.

As before, we will analyze the effects of shutting down the monetary policy response by looking at the response of total expenditure, as well as the expenditure of households based on housing tenure status and age. Looking at the total change in expenditure when only mortgage rates are allowed to move in Figure G.1, we see that changes in the mortgage rates do explain large part of expenditure change in the context of the model. However, once the effect of the shock dies out, households temporarily decrease their expenditure, slowly converging back to the initial steady-state (while not visible in the graph, the expenditure reaches back the original value at the end of the simulation).

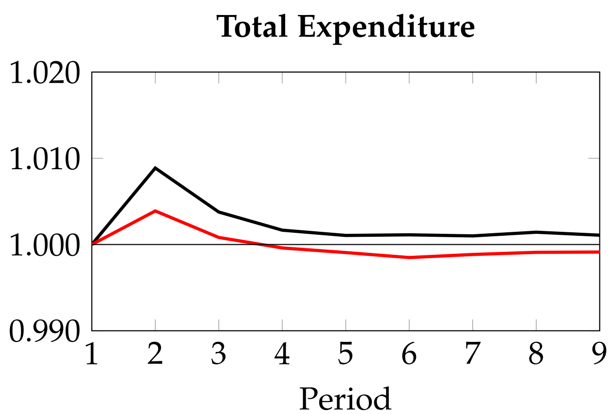

Figure G.1: The figure plots the simulated path of average expenditure for all households. Initial period is normalized to 1.

Since the only policy variable that is changing in this scenario are the mortgage rates, the households who still respond the strongest is the group of mortgagors (see top panel in Figure G.2), while the response of renters and outright owners is essentially flat (most

[5]The change is mortgage spreads alone is not only interesting to break down the transmission mechanism identified in the model. From the point of view of the monetary policy literature, for example, movements in spread is potentially important propagation mechanism of such policies. For example, Gertler and Karadi (2015) find that the spread between mortgage rates and short-term interest rates is a key reason why long-term rates respond to monetary policy shocks.

of the movements seen in the figure for those households is simulation error).

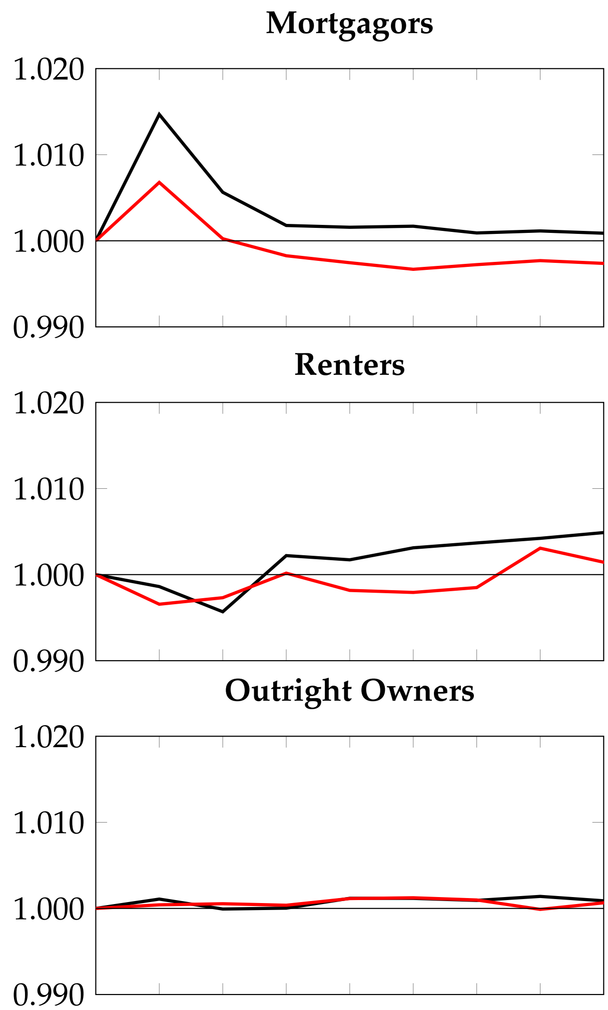

Figure G.2: The figure plots the simulated path of average expenditure for households based on the housing tenure status. Top panel is for mortgagors, middle panel is for renters, while bottom panel is for outright owners. Black line corresponds to the original experiment in Section 5, red line corresponds to the alternative experiment in the current section. Initial period is normalized to 1.

Finally, along the age dimension, as most of the younger households and large portion of the middle-aged households are exposed to mortgage debt, this group of households benefits the most from the decrease in the mortgage rates and they increase their expenditure even in the alternative experiment. Again, as in the case of overall expenditure response, the peak increase is now lower, given that only the mortgage rates change and this change is not as large anymore.

Figure G.3: The figure plots the simulated path of average expenditure for households based on their age. Top panel is for young households, middle panel is for middle-aged, while bottom panel is for older ones. Black line corresponds to the original experiment in Section 5, red line corresponds to the alternative experiment in the current section. Initial period is normalized to 1.

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